

# Job Description — Full-Time Position

C-DAC is seeking engineers passionate about developing next-generation computing technologies spanning hardware systems and advanced software platforms contributing to national technology development initiatives.

**Role:** Chip Design & System Software

**Employment Type:** Full-Time

**Organization:** Centre for Development of Advanced Computing (C-DAC)

**Location:** C-DAC Centre across India — **Bengaluru, Thiruvananthapuram, Pune, Hyderabad, Noida, Guwahati**

## Eligibility

### Academic Qualification

- **B.Tech:** Final year students (2027 passout)
- **M.Tech:** Final year students (2027 passout)
- **PhD:** Candidates who have completed thesis submission

### Eligible Branches (or equivalent disciplines)

Candidates from the following or related specializations may apply:

- Computer Science & Engineering (CSE)
- Electronics & Communication Engineering (ECE)
- Electrical & Electronics Engineering (EEE)
- VLSI Design / Microelectronics
- Embedded Systems
- Information Technology (IT)
- Computer Engineering
- Electronics Engineering
- Artificial Intelligence / Data Science
- Or any related discipline relevant to chip design or system software

## Responsibilities

### Chip Design Engineers

- Digital system and Microprocessor design
- RTL design and digital system development
- Functional simulation, verification and debugging
- Performance, power and area optimization
- FPGA prototyping and validation
- Collaboration with software teams for hardware–software co-design
- Support architecture evaluation and design exploration

## **System software Engineers**

- Design and development of system software components including compilers, debuggers, profilers
- Design and implementation of software toolchains and low-level programming utilities
- Performance analysis and optimization using profiling and debugging tools
- Development and optimization of mathematical and numerical libraries for high-performance computing
- Software development targeting CPUs, GPUs, and AI accelerator platforms
- Development of runtime support for efficient execution and resource management
- Optimization of software stacks considering processor architecture, memory hierarchy, and instruction sets
- Implementation of performance optimizations for heterogeneous and accelerator-based systems
- Optimization and integration of AI/ML workloads on heterogeneous computing platforms

## **Required Skills**

### **Common Skills**

- Strong analytical and problem-solving ability
- Understanding of computer systems fundamentals
- Familiarity with Linux development environments

### **Chip Design Track**

- Digital Electronics fundamentals
- Computer Architecture concepts
- RTL design using Verilog/SystemVerilog/VHDL
- Simulation and debugging methodologies
- Understanding of processor pipelines and memory systems

### **System Software Track**

- Programming in C/C++ and Python
- Data Structures & Algorithms
- Operating Systems fundamentals, especially Linux
- Knowledge of compiler concepts including IR, optimization, and code generation
- Understanding of computer architecture and ISA
- Familiarity with system-level debugging and performance analysis tools
- Basic understanding of AI/ML workloads, frameworks, or model execution pipelines